

# SHUVAM PAL

☎ +91 85830 54679 ✉ ishuvam.pal@gmail.com 🌐 Shuvam Pal 🕒 unichronic

## Education

Dayananda Sagar College of Engineering

Bengaluru, India

*B.E. Artificial Intelligence and Machine Learning*

## Experience

### Cent Health

Jul 2025 – Present

*Machine Learning Intern: Python, Medical AI, ONNX, TinyGrad, Kafka, Redis, AWS*

Bengaluru

- Built and deployed a scalable **ML pipeline** for **organ segmentation** and organ-level quality control, flagging **physical abnormalities** and implausible measurements in patient organs.
- Built an **evaluation pipeline** for the company's segmentation and report-verification models, with repeatable checks to measure and review model quality over time.
- Built a **brain-age model from scratch** using **T1-weighted MRI** scans. Used **SPM12** to prepare the scans and extract grey-matter maps, **PCA** to reduce the number of features, and **Gaussian Process Regression** to predict age.
- Reduced the model's tendency to pull predictions toward the average age with a small **age-calibration correction** around the average training age, tuned on **held-out validation folds**; added explanations showing which MRI feature patterns influenced each estimate and checked errors and uncertainty across age groups and on a separate dataset.
- Used deep-learning models for **liver and kidney segmentation** and automated **liver span** and **renal cortical thickness** measurements.
- Improved the in-house **PACS image-loading time** by profiling the **DICOM-series path** across metadata parsing, pixel-data decoding, preprocessing, and viewer rendering, then reducing repeated work before display.

### Bentham AI

Dec 2025 – Mar 2026

*Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions*

Remote

- Built a **Puppeteer/Node.js automation tool** for MCA company registrations, with API lookups, retries, session recovery, and validation checks; cut manual filing time by **85%**.
- Added structured draft generation for company names, descriptions, and filing fields, with inspectable outputs and validation before submission.
- Packaged Go/Node.js services with **Docker**, deployed them to **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment speed by **70%**.

### Google Summer of Code

May 2025 – Sept 2025

*Mentee at Invesalius: Python, VTK, NumPy, PyTorch, ONNX, TinyGrad*

Remote

- Built a **brain MRI segmentation workflow** for **95 anatomical regions**, adding NIFTI/MGZ preprocessing, label mapping, mask generation, and quick **QC checks** for model outputs.
- Added **PyTorch and ONNX** model inference to InVesalius, using **multiprocessing** and **memory-mapped volumes** to cut processing time by **87%**.
- Converted segmentation label volumes into **LUT-based masks** and **VTK PolyData** surfaces using **marching-cubes reconstruction** for 3D visualization, measurement, and export; preserved affine image-to-world transforms for neuronavigation and tracked targets.
- Extended InVesalius's marker system for **TMS-coil neuronavigation**, creating **stimulation-point markers** from tracked coil poses and updating their positions in MRI slices and the 3D view in real time.

## Projects

**Postificus Content Distribution Engine** | *Go, Echo, PostgreSQL, Redis, RabbitMQ, Go-Rod*

[GitHub](#) 

- Solved fragmented publishing across Medium, Dev.to, and LinkedIn with a centralized **Go/Echo engine** built around **Go-Rod browser automation** and AI-driven content transformation, turning one draft into platform-ready posts.
- Engineered distributed execution for long-running browser jobs with **RabbitMQ retries and dead-letter queues**, **Redis** autosave, **PostgreSQL** persistence, circuit breakers, health checks, and **Prometheus** metrics to isolate third-party failures.

## Technical Skills

**Languages:** Python, Go, JavaScript, TypeScript

**Frameworks:** FastAPI, Echo, Node.js, PyTorch

**Tools:** PostgreSQL, Redis, RabbitMQ, Docker, GCP, GitHub Actions, ONNX, SPM12, VTK, DICOM/PACS